

Melanie Escobar Marulanda

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EDUCATION

University of Houston

Bachelor of Science in Computer Science | GPA: 3.92/4.00

Houston, TX

May 2026

PROJECTS

IEEE-NSM Website – www.ieee-nsm.com

July 2025 – Present

- Led a 3-member team to develop IEEE-NSM's website using **React** and **CSS**, deployed with **Vercel**, reaching over 100 members.
- Integrated **Google Sheets**, **Calendar**, and **Apps Script** to handle event scheduling, image uploads with real-time updates.
- Streamlined officer content management by automating dynamic updates through Google Apps Script backend logic.

FoodLens – iOS Nutrition Tracker

Oct 2025 – Present

- Designed UI with **Figma** and developed frontend with **Swift** and **Xcode** to help users log meals, set goals, and track nutrition.
- Built a Home, Trends, and Settings view with **SwiftUI** for meal logging, macro visualization, and personalized goal management.
- Currently integrating **Core Data** and **machine learning**-based food recognition to enhance smart tracking.

Drug Consumption Analysis – Machine Learning, Data Science & Statistical Learning

Oct 2025 – Present

- Analyzed substance-use using UCI dataset with **Python**, **Pandas**, **NumPy**, **Scikit-Learn**, **TensorFlow**, and **Matplotlib**.
- Currently building supervised models with **KNN**, **SVM**, **Random Forest**, and **Neural Network**.
- Currently Investigating key demographic and personality traits influencing substance-use predictions.

Spa AI – AI Video Chatbot

Jan 2025 – May 2025

- Developed a full-stack AI chatbot for a real-world client, integrating **OpenAI** and **Gemini** via **LangChain** for intelligent video-based responses with seamless model switching through an admin UI.
- Engineered backend logic with **Python** and **FastAPI** using embeddings for product/context matching; Containerized the application with **Docker** and managed data persistence using **Supabase**.
- Achieved 100% code coverage (**Jest**, **Pytest**) and integrated **Jenkins CI/CD** for automated testing and deployment.

The Museum of Fine Arts, Houston – Full-Stack Management System

Aug 2024 – Nov 2024

- Developed a full-stack platform using **React**, **Node.js**, **Express.js**, and **MySQL** to manage museum collections, exhibitions, events, artwork, gift shop inventory, memberships, role-based authentication, reporting, and database triggers.
- Implemented full **CRUD functionality** with form validation and image hosting via **Azure Blob Storage**.
- Created reporting tools using **jsPDF** for generating downloadable ticket-sales summaries.

Shasta's Coogmunity Service – Full-Stack Volunteer Management System

Aug 2024 – Nov 2024

- Developed a volunteer-event matching platform using **React**, **Node.js**, **Express**, and **MongoDB**.
- Implemented event matching logic based on profiles (skills, location, and availability) with role-based authentication via **JWT**.
- Added PDF/CSV export features using **pdfkit** and **json2csv** for admin reporting.

LEADERSHIP AND WORK EXPERIENCE

Lead Webmaster Officer

IEEE-NSM

July 2025 – Present

University of Houston

- Built and launched the organization's website
- Coordinated with officers and committees to publish event and streamline content updates

Academics Officer

ColorStack UH Chapter

July 2025 – Present

University of Houston

- Organized **LeetCode practice sessions** designed to provide interview prep practice for chapter members.
- Created a centralized hub containing academic resources using **Notion**.

Crew Member

Chipotle

August 2024 – August 2025

Pearland, TX

- Delivered professional, friendly service while maintaining high customer satisfaction and positive feedback from managers.
- Prepared and assembled orders accurately and efficiently on the front line, ensuring strict adherence to food safety standards in a fast-paced environment.